WORKED SOLUTION

STEP III 1991, Question 1

Pure • Series and Expansions

STEP III 1991, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q1

The key idea

All three parts reward a moment's thought before any grinding. In (i), partial fractions turn the sum into pieces that telescope. In (ii), the cubic $1+x+x^2+x^3$ is a geometric series, so it is a quotient of two very simple expressions, and the logarithm splits into two standard series. In (iii), write the exponent as a series first; since it starts at x^2, only a couple of terms of the exponential series can reach x^4.

Solution

Part (i)

We look for constants with

$$\frac{6}{r(r+1)(r+3)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+3}.$$

By the cover-up rule, $A = \dfrac{6}{1\cdot 3} = 2$, $B = \dfrac{6}{(-1)(2)} = -3$ and $C = \dfrac{6}{(-3)(-2)} = 1$. (As a quick check, $A+B+C=0$, as it must be because the left-hand side behaves like $6/r^3$ for large r, with no $1/r$ term.)

The coefficients $2, -3, 1$ sum to zero, which suggests splitting them as $2-2$ and $-1+1$:

$$\frac{6}{r(r+1)(r+3)} = 2\left(\frac{1}{r} - \frac{1}{r+1}\right) - \left(\frac{1}{r+1} - \frac{1}{r+3}\right).$$

Each bracket now telescopes. The first gives

$$\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+1}\right) = 1 - \frac{1}{n+1}.$$

In the second, the terms are two steps apart, so two terms survive at each end:

$$\sum_{r=1}^{n}\left(\frac{1}{r+1} - \frac{1}{r+3}\right) = \frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3}.$$

Putting these together,

$$\sum_{r=1}^{n} \frac{6}{r(r+1)(r+3)} = 2 - \frac{2}{n+1} - \frac{5}{6} + \frac{1}{n+2} + \frac{1}{n+3}$$
$$= \frac{7}{6} - \frac{2}{n+1} + \frac{1}{n+2} + \frac{1}{n+3}.$$

Part (ii)

? **Ask yourself** Is there a neater way to write $1 + x + x^2 + x^3$?

It is a geometric series with ratio x, so for $x \neq 1$

$$1 + x + x^2 + x^3 = \frac{1 - x^4}{1 - x}.$$

For $|x| < 1$ both $1 - x^4$ and $1 - x$ are positive, so we may split the logarithm:

$$\ln(1 + x + x^2 + x^3) = \ln(1 - x^4) - \ln(1 - x).$$

Using $\ln(1 - y) = -\sum_{k \geq 1} y^k / k$, valid for $|y| < 1$ (and $|x^4| < 1$ here too),

$$-\ln(1 - x) = \sum_{k=1}^{\infty} \frac{x^k}{k}, \qquad \ln(1 - x^4) = -\sum_{m=1}^{\infty} \frac{x^{4m}}{m}.$$

The second series only touches powers x^k with k a multiple of 4. For such $k = 4m$ the combined coefficient is

$$\frac{1}{4m} - \frac{1}{m} = -\frac{3}{4m} = -\frac{3}{k}.$$

So

$$\ln(1 + x + x^2 + x^3) = x + \frac{x^2}{2} + \frac{x^3}{3} - \frac{3x^4}{4} + \frac{x^5}{5} + \cdots,$$

and the general term is

$$\frac{x^k}{k} \text{ if } 4 \nmid k, \qquad -\frac{3x^k}{k} \text{ if } 4 \mid k.$$

Every coefficient is non-zero, so the five terms displayed are the first five non-zero terms.

Part (iii)

For $-1 < x \leq 1$ we have $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so

$$u = x \ln(1 + x) = x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \cdots.$$

? **Ask yourself** How many terms of $e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \cdots$ do we need?

Because u starts at x^2, the term u^k starts at x^{2k}. So u^3 and beyond contribute nothing up to x^4, and from u^2 we need only its leading term x^4. Hence

$$\begin{aligned} e^{x\ln(1+x)} &= 1 + \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3}\right) + \frac{1}{2}x^4 + \cdots \\ &= 1 + x^2 - \frac{x^3}{2} + \frac{5x^4}{6} + \cdots . \end{aligned}$$

(Of course $e^{x\ln(1+x)}$ is just $(1+x)^x$.)

Checks

- In (i), $n = 1$ gives $\frac{7}{6} - 1 + \frac{1}{3} + \frac{1}{4} = \frac{3}{4}$, which matches the single term $\frac{6}{1\cdot 2\cdot 4}$. Letting $n \to \infty$ gives the sum to infinity $\frac{7}{6}$.
- In (ii), expanding directly to second order, $\ln(1+y) \approx y - \frac{y^2}{2}$ with $y = x + x^2 + \cdots$, gives $x + x^2 - \frac{x^2}{2} = x + \frac{x^2}{2}$, agreeing with our first two terms. Numerically, at $x = 0.5$ the series summed to many terms agrees with $\ln 1.875 \approx 0.6286$.
- In (iii), at $x = 0.01$ the expansion differs from $(1.01)^{0.01}$ by about 7.4×10^{-11}, which is of size x^5, as it should be (the next coefficient turns out to be $-\frac{3}{4}$).

Common slips

- In (i), forgetting that a telescoping sum with a gap of two leaves two terms at each end, not one.
- In (ii), giving the coefficient of x^4 as $\frac{1}{4} - 1$ but then writing the general term as $(-1)^{k+1}x^k/k$ or similar; the pattern depends on k modulo 4.
- In (iii), dropping the $\frac{1}{2}u^2$ term: it contributes $\frac{1}{2}x^4$.

Beyond STEP

Part (iii) is a small example of composing power series, and the bookkeeping trick of tracking the lowest power in each term is exactly how asymptotic expansions are handled at university. The questions of when a power series converges, and why one can add, multiply and compose them freely inside the radius of convergence, are the business of a first course in analysis; Abbott's *Understanding Analysis* treats them clearly.

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